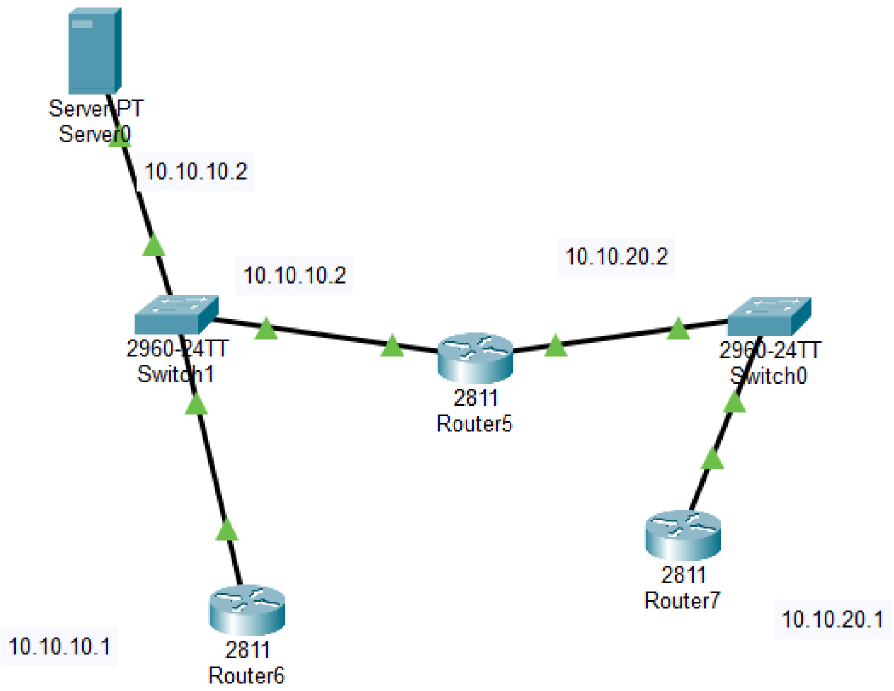
Week 3 practical

Semester 2

Topology




```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#no ip address
Router(config-if)#ip address 10.10.20.1 255.0.0.0
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#
```

Question 16

Router 5

```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip domain-lookup
Router(config)#ip name-server 10.10.10.10
Router(config)#ip domain-name lab.local
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.10.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/7 ms

Router#
```

Router6

```
Router#
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip domain-lookup
Router(config)#ip name-server 10.10.10.10
Router(config)#ip domain-name lab local
                                    ^
% Invalid input detected at '^' marker.

Router(config)#ip domain-name lab.local
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 7

```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip domain-lookup
Router(config)#ip name-server 10.10.10.10
Router(config)#ip domain-name lab.local
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.10.20.1
```

Question 17

```
Router#ping 10.10.20.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/2/7 ms

Router#
```

Question18

```
Router>en
Router#clear arp-cache
Router#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/1/3 ms

Router#
```

Question 19

Rouer 6

```
Router#
Router#
Router#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.10.10.1              -   00D0.9798.A501  ARPA   FastEthernet0/0
Internet  10.10.10.2              2   0000.0CBE.1501  ARPA   FastEthernet0/0
Router#
```

Router 5

```
Router#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.10.10.1              3   00D0.9798.A501  ARPA   FastEthernet0/0
Internet  10.10.10.2              -   0000.0CBE.1501  ARPA   FastEthernet0/0
Internet  10.10.10.10            41   0060.704A.4C8D  ARPA   FastEthernet0/0
Router#
```

Router 7

```
Router>
Router>show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  10.10.20.1              -   000B.BEAE.CC01  ARPA   FastEthernet0/0
Router>
```

Question 20

An (ARP) entry provides the physical hardware or MAC address of a specific device on the local network. A routing table entry only tells a device the next-hop IP address, network subnet, or exit interface needed to forward a packet toward its destination, but it does not contain hardware addresses.

Question 21

Because IP routing and Ethernet delivery are 2 different jobs